Ethan Villalovoz

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Education

Georgia Institute of Technology — Expected May 2028
Master of Science in Computer Science (Computational Perception & Robotics) | GPA: 4.0/4.0 — *Atlanta, GA*

Washington State University — Aug 2021 – May 2025
Bachelor of Science in Computer Science, Minor in Mathematics | GPA: 3.94/4.0 — *Pullman, WA*

Experience

Georgia Institute of Technology — Aug 2026 – Present
Graduate Student Researcher — *Atlanta, GA*

- Investigating action-conditioned world models and test-time adaptation for robot learning.

Microsoft — May 2026 – Jul 2026
Software Engineer Intern — *Redmond, WA*

- Built a **C#/.NET AI-agent prototype** using GraphQL to investigate financial-close blockers across applications.
- Built a 36-question evaluation harness to guide model and prompt changes, cutting inference cost about **30×**.
- Replaced individual record lookups with an indexed query, reducing database round trips from **201 to 1 per scan**.
- Lazy-loaded React routes and a chat widget, reducing the application's initial bundle size by **39.5%**.
- Fixed **22 security findings** in two production repositories, including file-access and Azure configuration flaws.

Washington State University — Jan 2024 – May 2025
Undergraduate Research Assistant — *Pullman, WA*

- Built an evaluation pipeline to test LLM-generated code on **164 HumanEval+ tasks across 3 models**.
- Benchmarked **Bayesian prompt optimization** against OPRO: **11–16 percentage-point gains** in tasks solved.

Carnegie Mellon University — Jun 2024 – Aug 2024
Robotics Institute Summer Scholar — *Pittsburgh, PA*

- Implemented **Bayesian reward learning** in Python to infer object placement preferences from corrections.
- Added questions about object color, type, and material to distinguish competing reward models in simulation.

Google — May 2023 – Aug 2023
Software Engineering Intern (STEP) — *Sunnyvale, CA*

- Developed **5 C++/SQL jobs** to collect internal database queue statistics across the full dataset **within 4 hours**.
- Built **HTML dashboards with SQL-like queries** for internal teams to monitor queue statistics and job results.

Publications

An Exploratory Study of Bayesian Prompt Optimization for Test-Driven Code Generation with Large Language Models. S. Tomar, A. Deshwal, **E. Villalovoz**, M. Fazzini, H. Cai, J. R. Doppa. *arXiv preprint, 2025*

Social Triangles and Aggressive Lines: Multi-Robot Formations Impact Navigation and Approach. A. Bacula, **E. Villalovoz**, D. Flynn, A. Mehta, H. Knight. *IEEE/RSJ IROS, 2023*

Open-Source Contributions

MTEB | Embedding Evaluation — PRs: #5321, #5266

- Added final-score regression checks across three integration suites to detect changes in benchmark results.
- Repaired two retrieval benchmarks by removing a corrupt image and validating 6,991 preserved images.

Hugging Face LeRobot | Robot Learning — PRs: #4418, #4799

- Corrected robot recording and training tutorials, fixing broken setup instructions and dataset mismatches.

Projects

RAG with Knowledge Graphs and Vector Search | Python, React, FastAPI, FAISS

- Built a RAG app using FAISS, DBpedia, and GPT-4o mini for Q&A over **10,000+ Wikipedia articles**.
- Automated 16 tests with pytest and GitHub Actions to check NLP, retrieval, and response generation.

Technical Skills

Languages: Python, C++, C, C#, SQL, TypeScript, JavaScript, Bash, PowerShell, HTML/CSS, MATLAB, R, Swift
Frameworks/Libraries: PyTorch, JAX, Transformers, scikit-learn, NumPy, pandas, React, FastAPI, .NET, OpenCV
Tools/Platforms: Git, Linux, Docker, GitHub Actions, Azure, AWS, MySQL, Prometheus, TensorRT, MLflow, ROS